

6.2410 Design Project – Spring 2026

Cavity-Enhanced Pulsed Coherent Control and Readout of Room-Temperature NV Spin Ensemble

Applications for Quantum Magnetometry

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Magnetic fields are important but...

Conventional magnetometers:

- Need cryogenic cooling
- Bulky shielding
- Sacrifice sensitivity for size



Helium vector magnetometer (HVM) of the [Pioneer 10](#) and [11](#) spacecraft
Source: Wikipedia

Solid-state quantum sensors promise to change that!

AGENDA

1 Background & Approach

2 Experimental Methods

3 Results & Analysis

4 Conclusions & Future

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1 Background & Approach

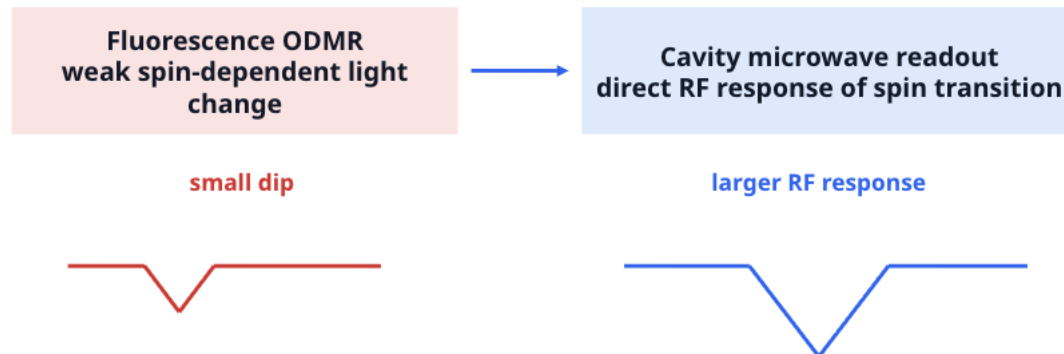
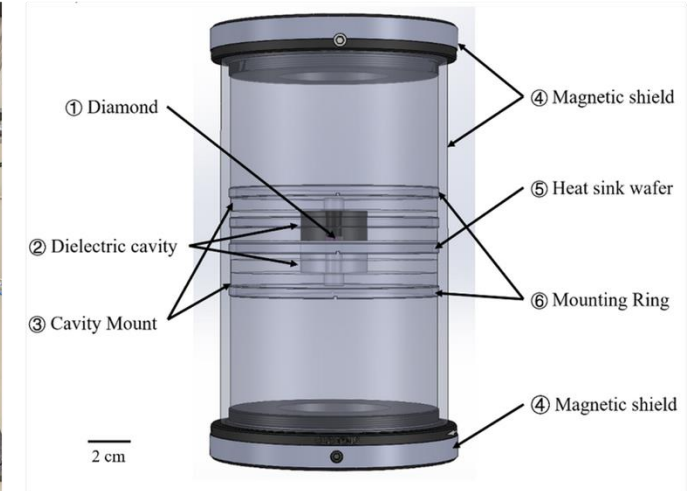
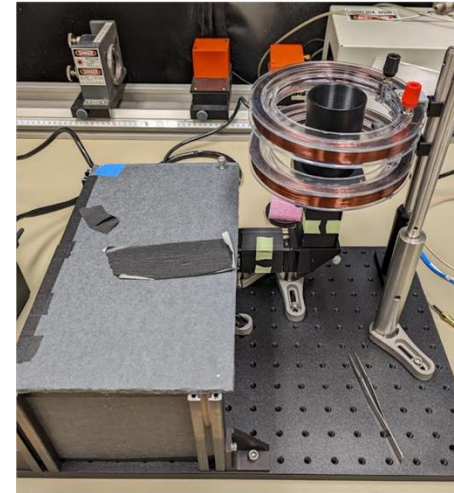
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High-Q cavities enhance NV readout for Room-Temperature Quantum Magnetometry but may distort pulsed spin control

- Pulsed time-domain control is essential for extracting T_2 , the spin coherence time that sets the fundamental sensitivity limit $\eta \sim 1/\sqrt{T_2}$, which CW frequency sweeps cannot access.
- NV centers in diamond are spin defects whose ground-state transitions are sensitive to magnetic field.
- A high-Q dielectric microwave cavity enhances interaction between the NV ensemble and microwave fields, => stronger microwave readout than conventional fluorescence-only detection.



- Implement pulsed coherent control on the cavity-enhanced NV platform
- Directly tests whether the high-Q cavity can also preserve microwave pulse fidelity for accurate spin manipulation w/ strong readout
- Controlling an NV ensemble rather than a single spin amplifies the collective coupling strength as $g\sqrt{N}$

NV spins are initialized optically, driven by microwaves, and read out through the cavity

- Ground-state spin transition:

$$m_s = 0 \leftrightarrow m_s = \pm 1$$

- Zero-field splitting:

$$D \approx 2.87 \text{ GHz}$$

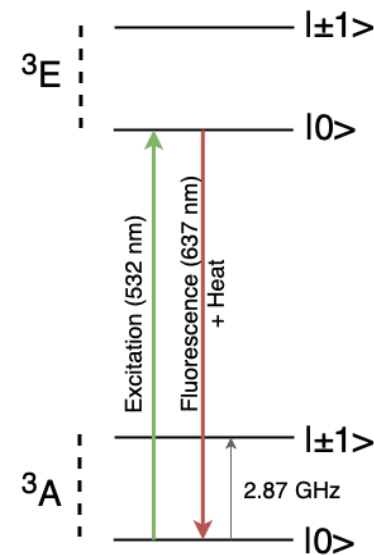
- Magnetic field causes Zeeman shift:

$$\omega_s(B) = D \pm \gamma B$$

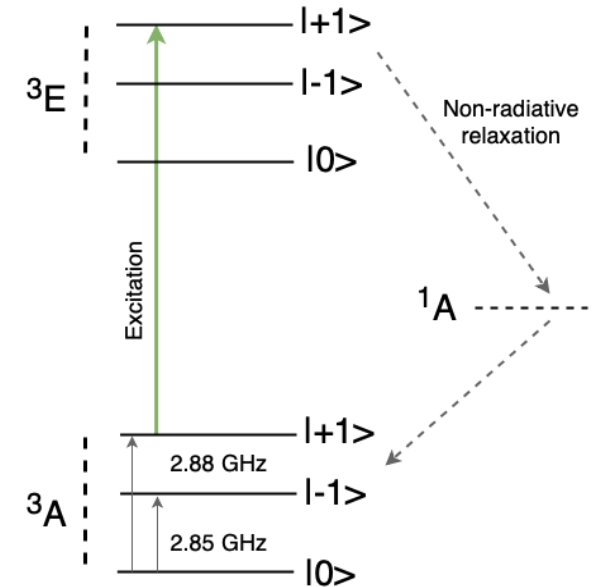
- Capabilities:

- optical initialization into $m_s = 0$ using a 532 nm laser
- microwave driving of spin transitions controlled by FPGA-timed microwave pulses under a static bias field B_0
- cavity-enhanced readout of spin response

No External Magnetic Field



External Magnetic Field



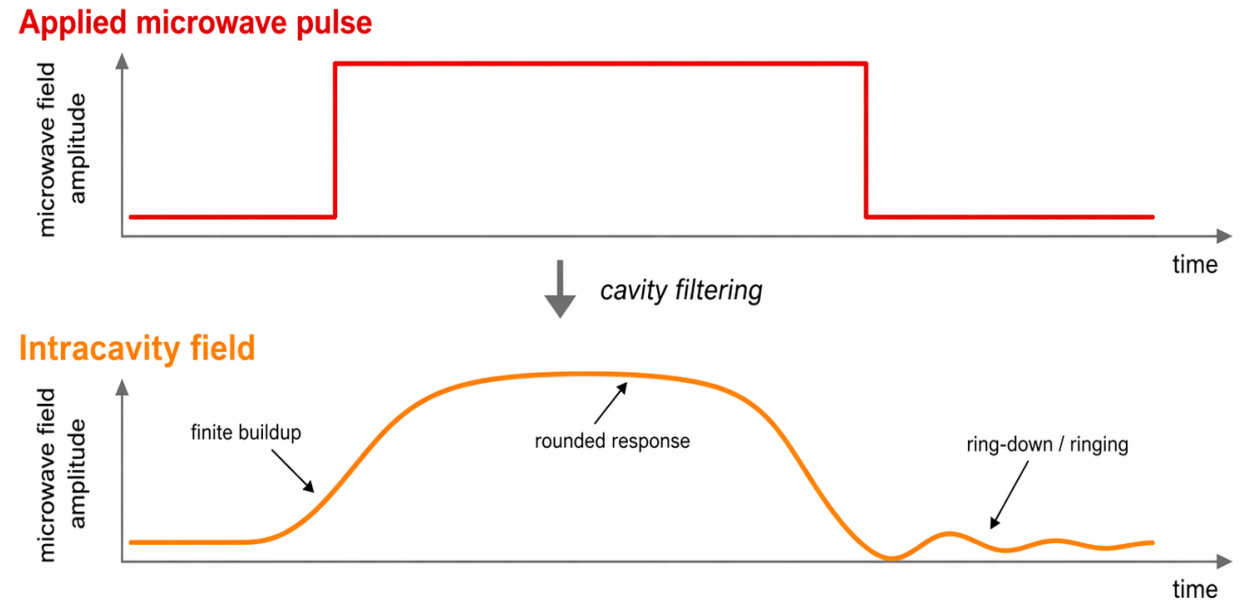
Ensemble control is critical: the collective coupling $g\sqrt{N}$ exceeds the single-spin coupling by $\sim\sqrt{N} \sim 10^3$, making spin-cavity interaction observable at room temperature without cryogenic amplification

Can High- Q Cavities Support Accurate Pulsed Control?

- High- Q cavities improve readout by sharpening the microwave response.
- But they also have **finite buildup and ring-down times**.
- Therefore, the intracavity field may differ from the applied rectangular pulse.
- This may distort measured Rabi oscillations and bias extracted parameters:

$$\Omega_R, T_R$$

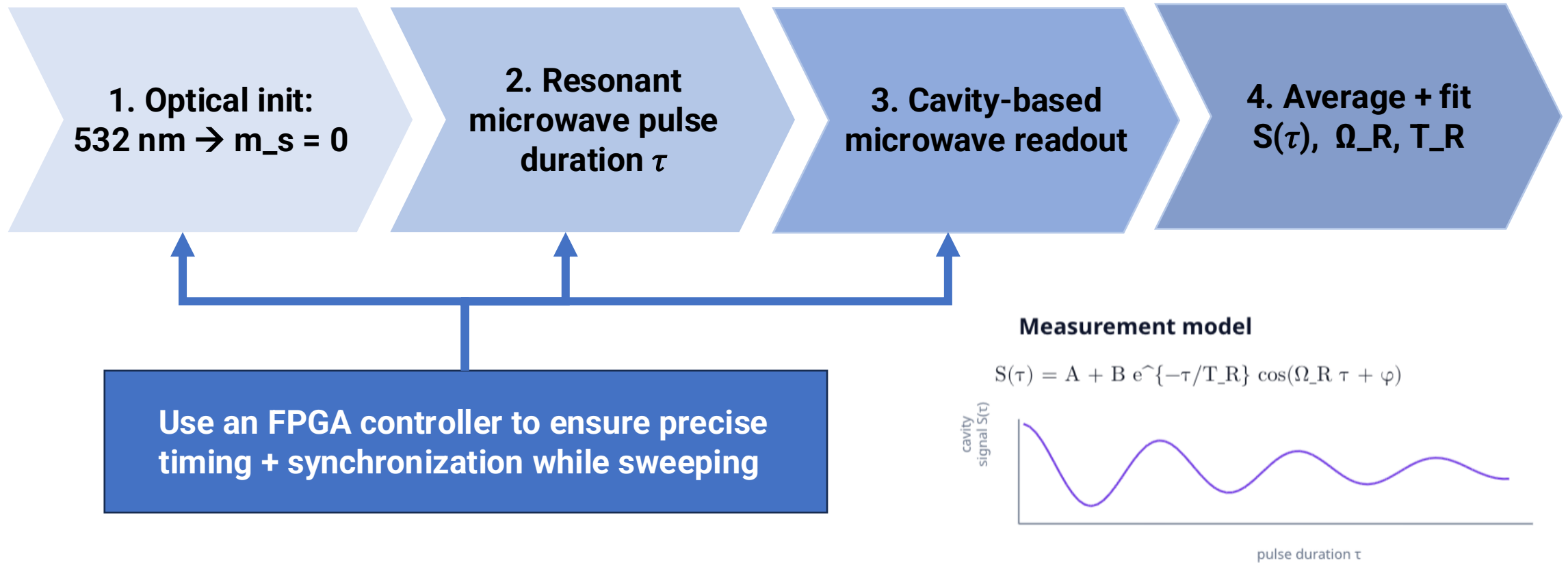
High- Q cavities reshape short control pulses



Spins see $B_1(t)$, not the ideal input waveform

Possible bias in extracted Ω_R and T_R

FPGA-timed pulse sequences sweep τ to extract Ω_R and T_R via cavity readout



→ Consider the differences between ideal vs distorted pulses

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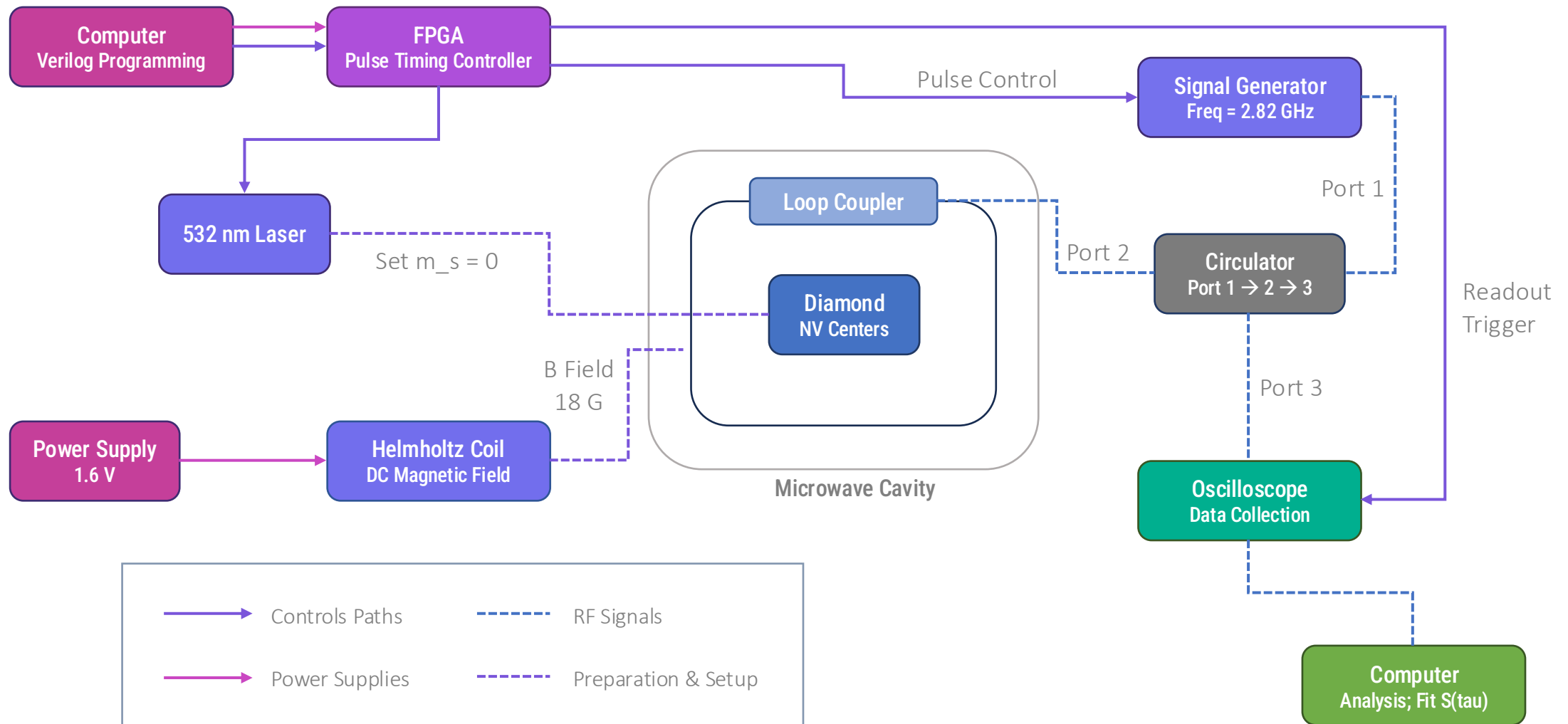
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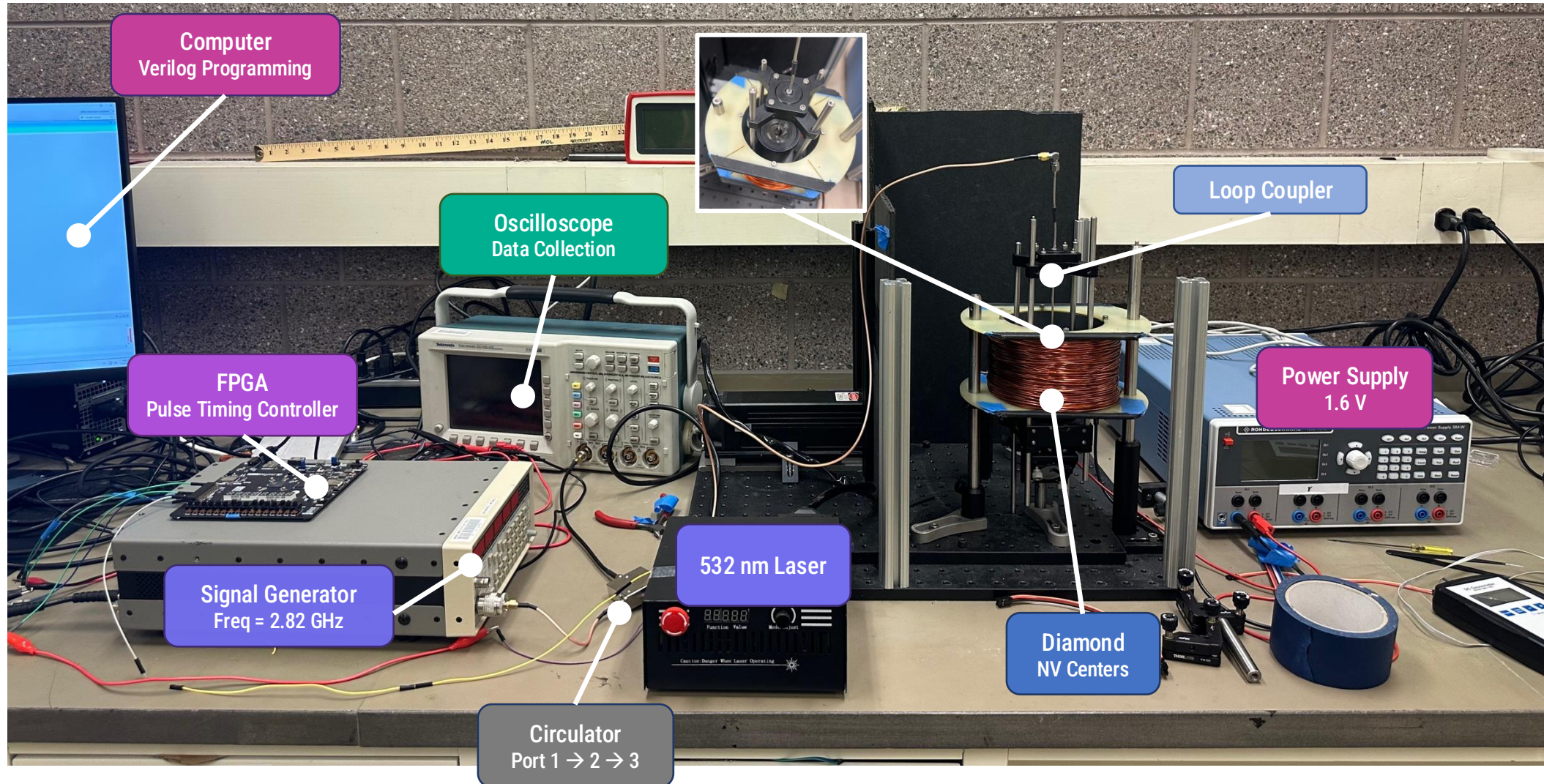
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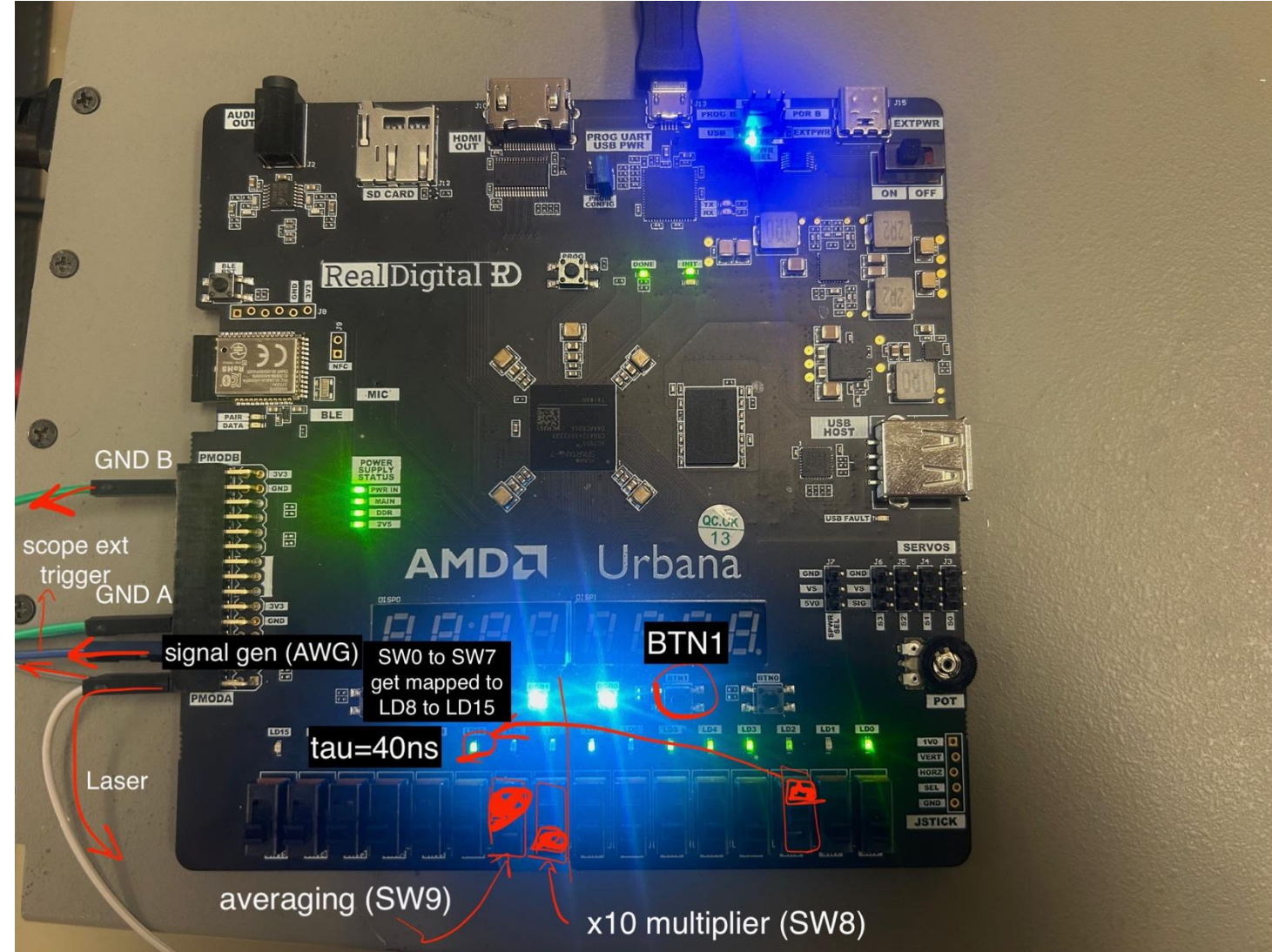
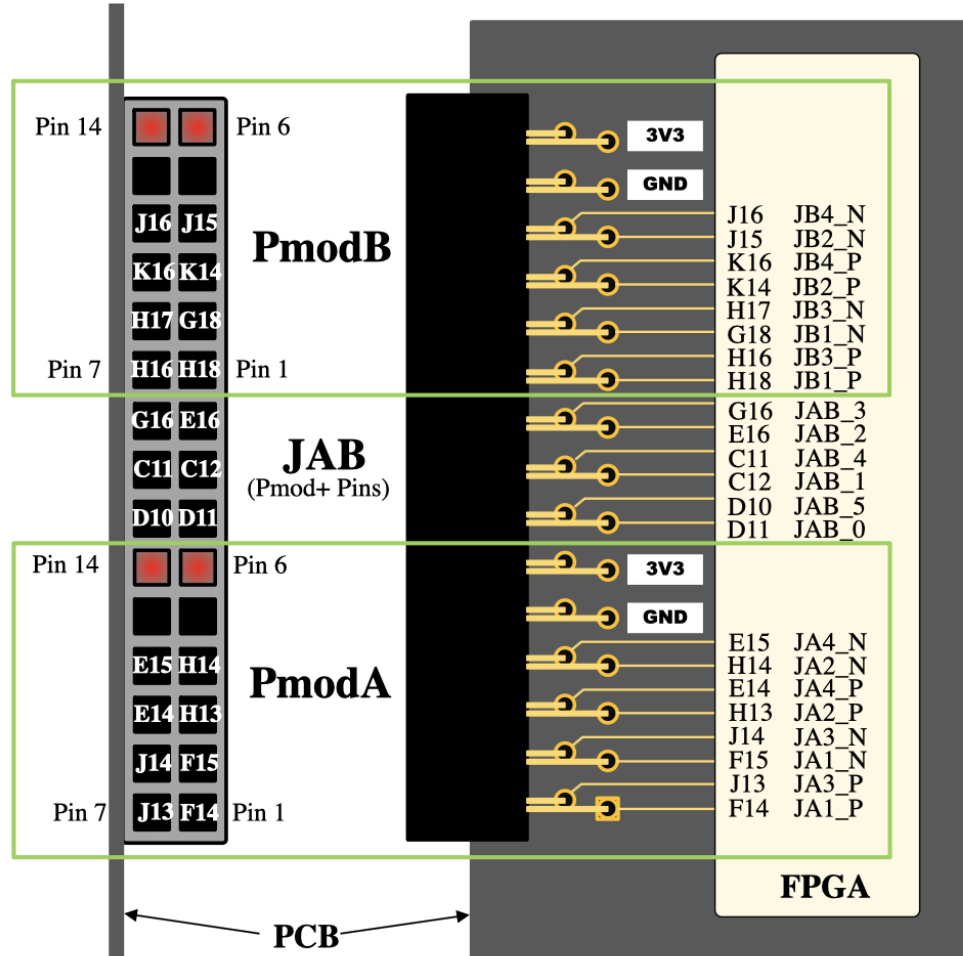
The FPGA synchronizes the laser, the microwave pulse, and the oscilloscope readout



The physical setup couples a diamond NV ensemble to a cylindrical dielectric resonator

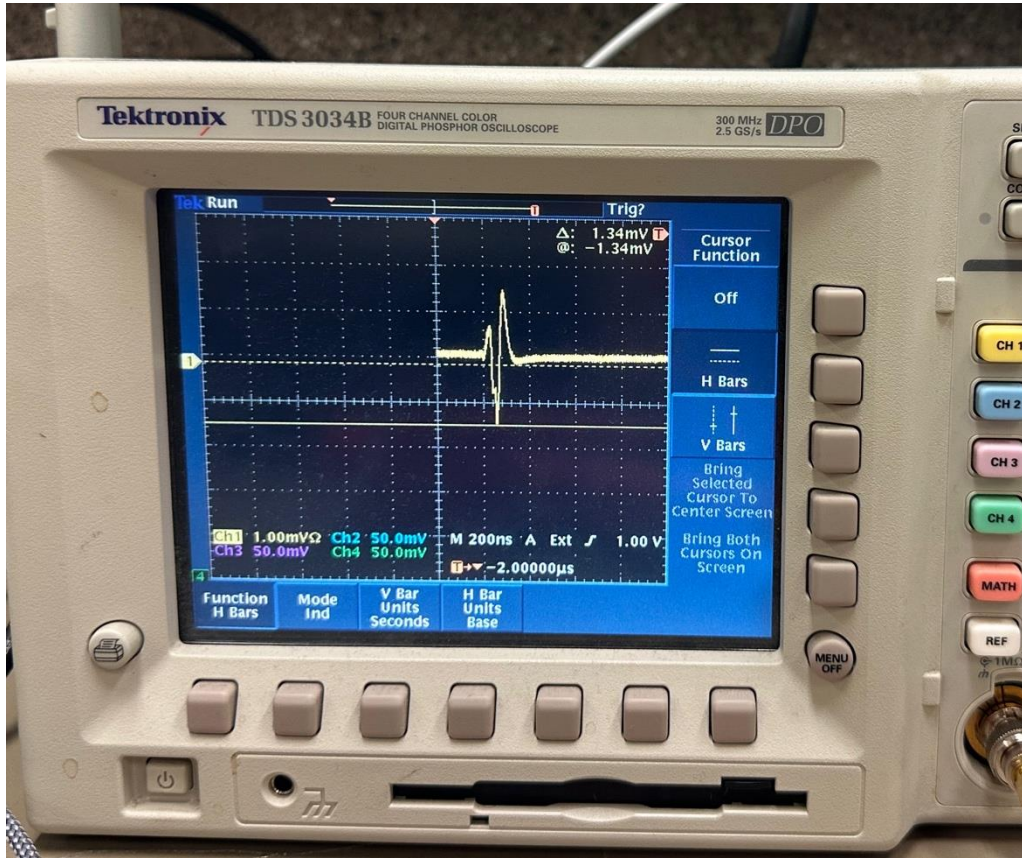


FPGA



Data is measured by manually reading the oscilloscope amplitude

The FPGA generates the full measurement sequence: 50 μ s laser polarization $\rightarrow$ 500 μ s settling $\rightarrow$ variable-duration MW pulse (τ) $\rightarrow$ readout trigger. Each press of BTN1 fires one sequence, or repeated sequences if in averaging mode.



For each τ value:

1. Set τ on the FPGA via DIP switches (SW0–SW7 encode τ in 10 ns steps; SW8 enables 10 $\times$ scaling)
2. Compile and load the Verilog design via Vivado, or simply change switches if already loaded
3. Connect the laser modulation input to the FPGA's laser_trig output (DB15 pin 15)
4. Enable auto-trigger mode (SW9 up) — the FPGA fires sequences continuously at $\sim$ 640/sec, allowing the scope to accumulate averaged traces rapidly
5. Set the oscilloscope to averaging mode (typically $N = 128$ or 512) and trigger on the FPGA's readout_trig (EXT input)
6. Wait $\sim$ 15 s for the average to converge
7. Use scope cursors to measure the amplitude of the cavity-reflection envelope (the second feature, at the falling edge of the MW pulse)
8. Repeat for the next τ value (no recompilation needed; switches alone determine τ)

* For testing purposes, we also ran experiments that skip steps 5 and 6 to consider the effects of taking a single pulse in comparison to the effects of taking the average.

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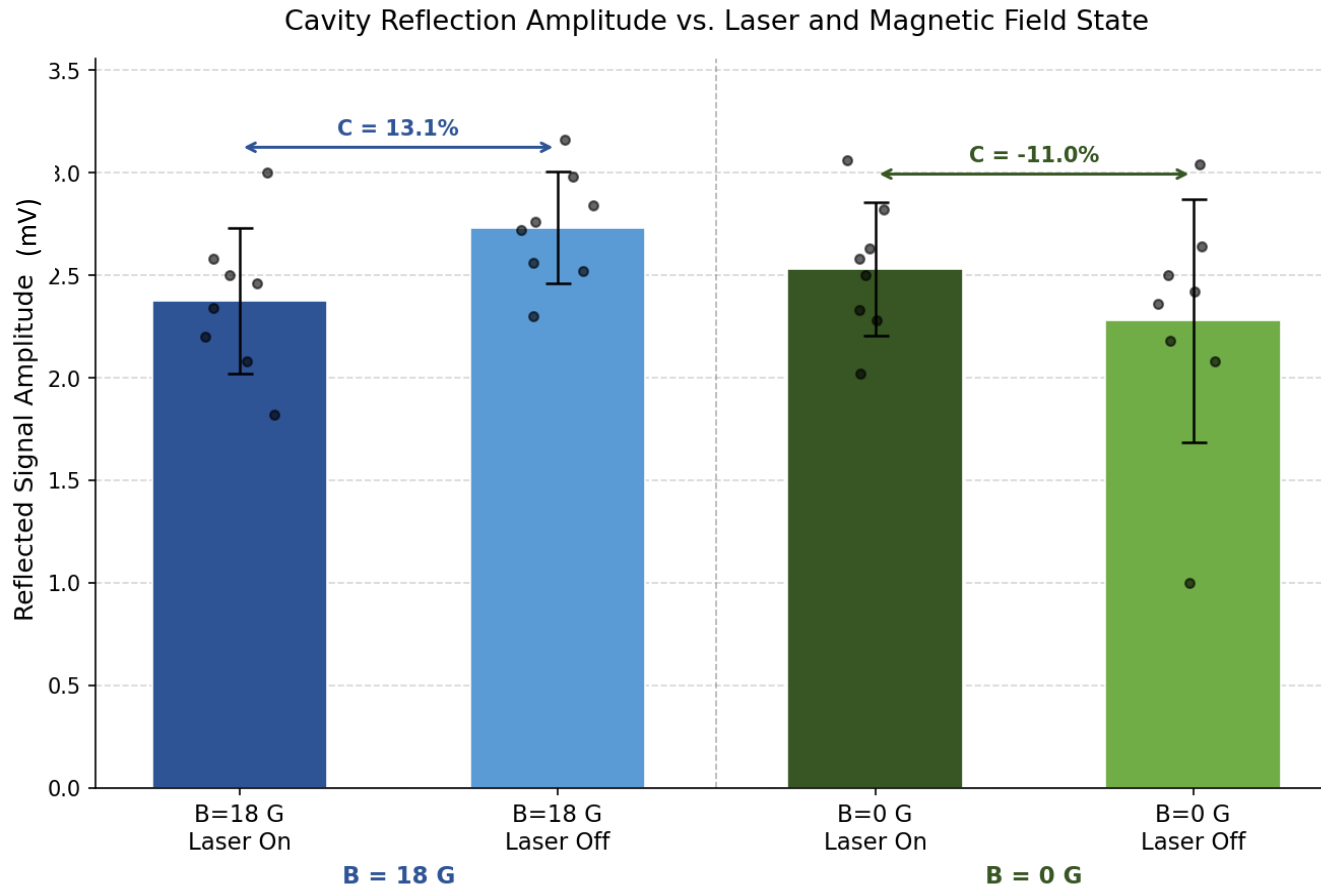
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Laser initialization and magnetic field tuning both contribute to measurable spin-cavity coupling

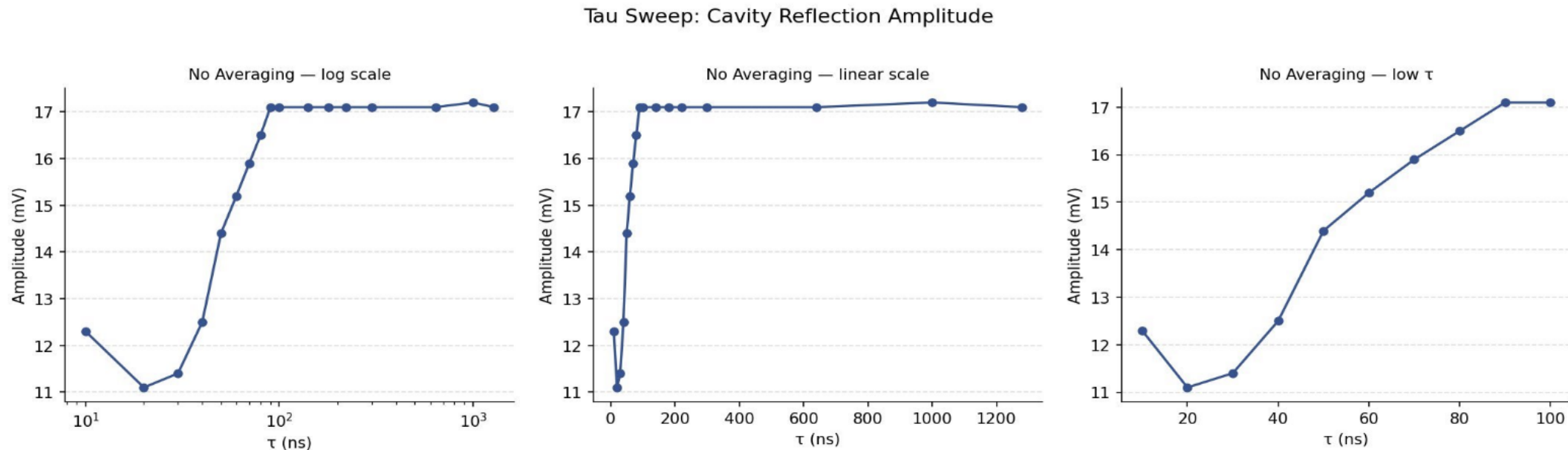


Additional Setup:

Tau = 1000 ns, Averaging is on for the Oscilloscope

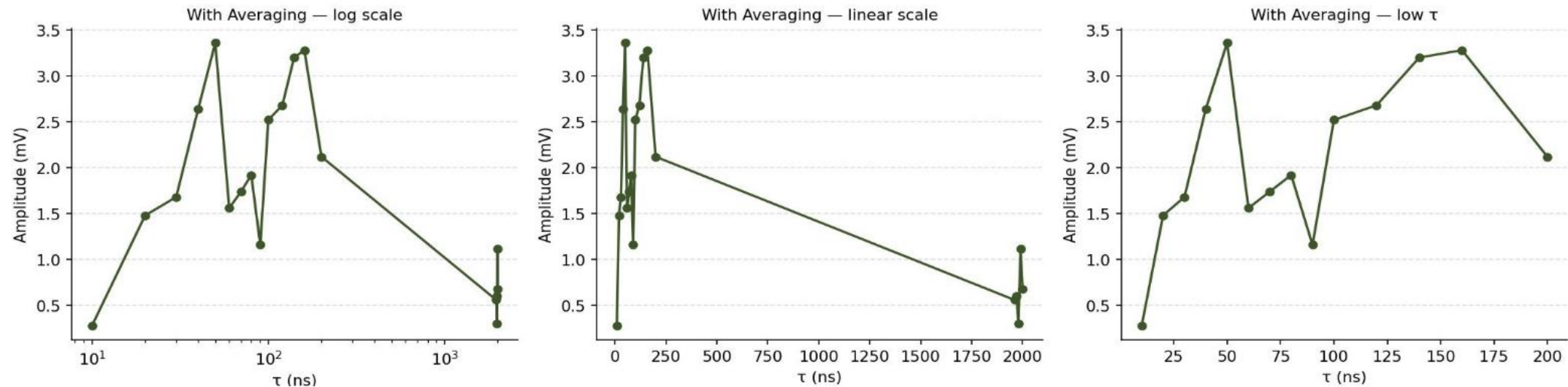
- $B = 18 \text{ G}$ gives 13% spin-dependent contrast, confirming the **NV ensemble is coupling to the cavity**
- **$B = 0 \text{ G}$ acts as a control**; contrast drops to $\sim -11\%$ and flips sign, confirming the $B = 18 \text{ G}$ signal is from spin-cavity coupling, not a laser heating artifact
- $B = 18 \text{ G}$ is our operating point for all subsequent pulse measurements
- **Noise is significant ($\sigma \approx 0.35 \text{ mV}$)**, motivating heavy averaging per τ point in the FPGA pulse sequence
- **13% cavity readout contrast exceeds the $\sim 2\text{--}5\%$ typical of optical fluorescence**, validating the cavity-enhanced readout architecture

The τ sweep shows cavity transient ringup at short τ and decay toward steady state at long τ ; coherent Rabi structure is below the noise floor



The amplitude rises monotonically from ~ 11 mV at $\tau = 20$ ns and saturates at ~ 17 mV by $\tau \approx 80$ – 100 ns, remaining flat through $\tau = 1280$ ns.

This is the cavity's transient response approaching its steady-state reflected power on the order of the cavity ringup time ($\sim 1/\kappa_c = 80$ – 100 ns), not Rabi dynamics.

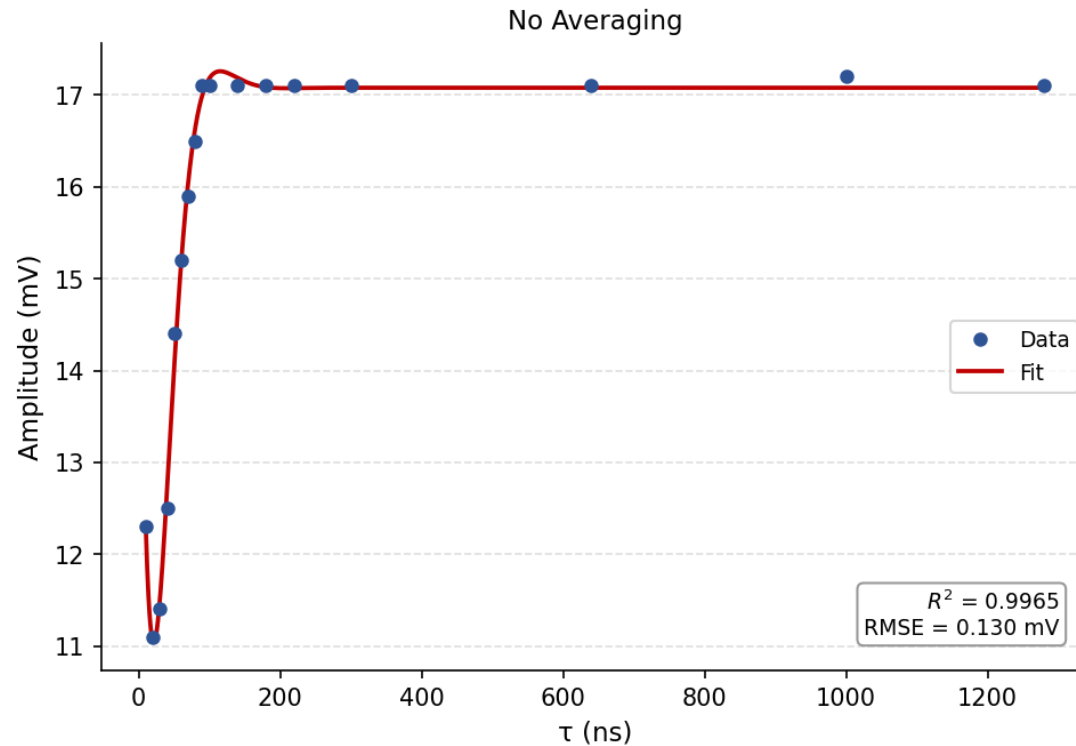


After noise reduction via $N = 512$ averaging, the signal drops to the mV range. The short- τ region (10–200 ns) shows scattered structure with no clearly resolvable oscillation, and the long- τ region (1960–2000 ns) shows a small steady-state value of ~ 0.6 mV.

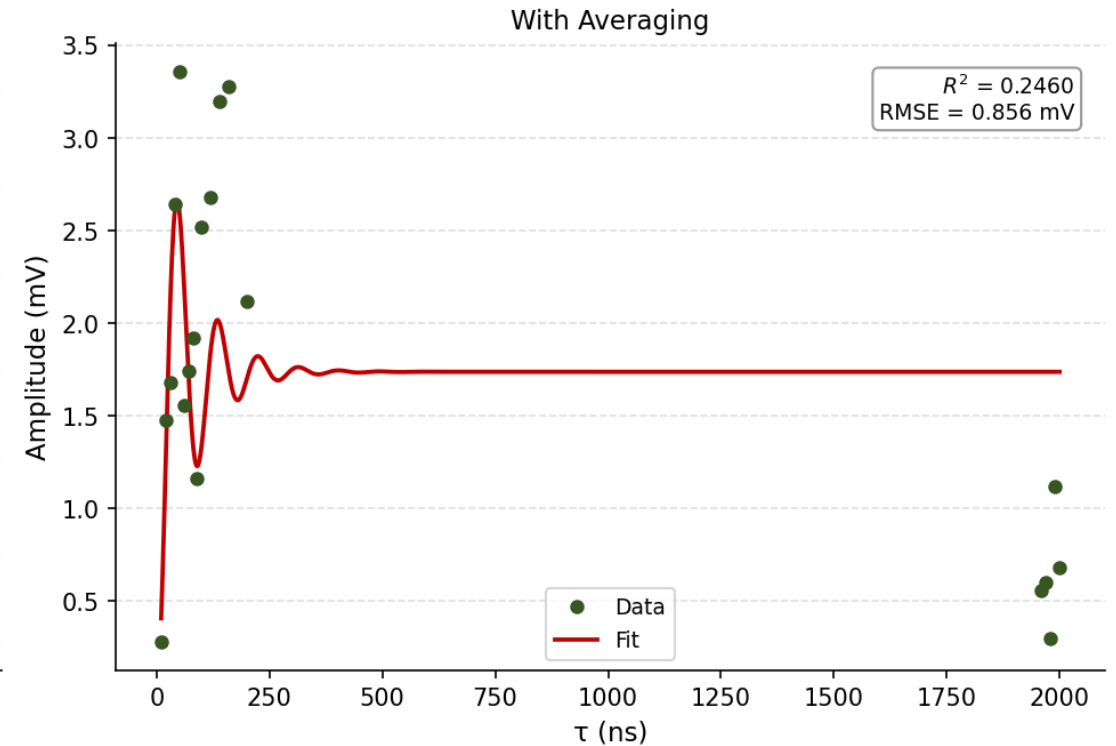
Interpretation: At short τ ($\tau < \sim 500$ ns), the trace is dominated by cavity dynamics rather than spin response. At long τ , the cavity has reached steady state but the NV contribution per shot is below the noise floor.

Extracting Ω_R and T_R via curve fitting proved to be unreliable

$$S(\tau) = A + B e^{-\tau/T_R} \cos(\Omega_R \tau + \phi) \text{ — Fit vs Data}$$



$$T_R = 26.74 \pm 2.48 \text{ ns};$$
$$\Omega_R = 33.423952 \pm 1.628477 \text{ rad/ns}$$



$$T_R = 74.72 \pm 92.95 \text{ ns}$$
$$\Omega_R = 70.211663 \pm 18.133326 \text{ rad/ns}$$

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We built the platform and detected coupling, but clean Rabi oscillations remain elusive

Key Results



Designed and validated a FPGA-based pulse sequencer with 10ns timing precision



Demonstrated pulsed cavity-NV measurement



Detected statistically significant NV-cavity coupling

Limitations



NV-cavity coupling is in the intermediate regime, not strong

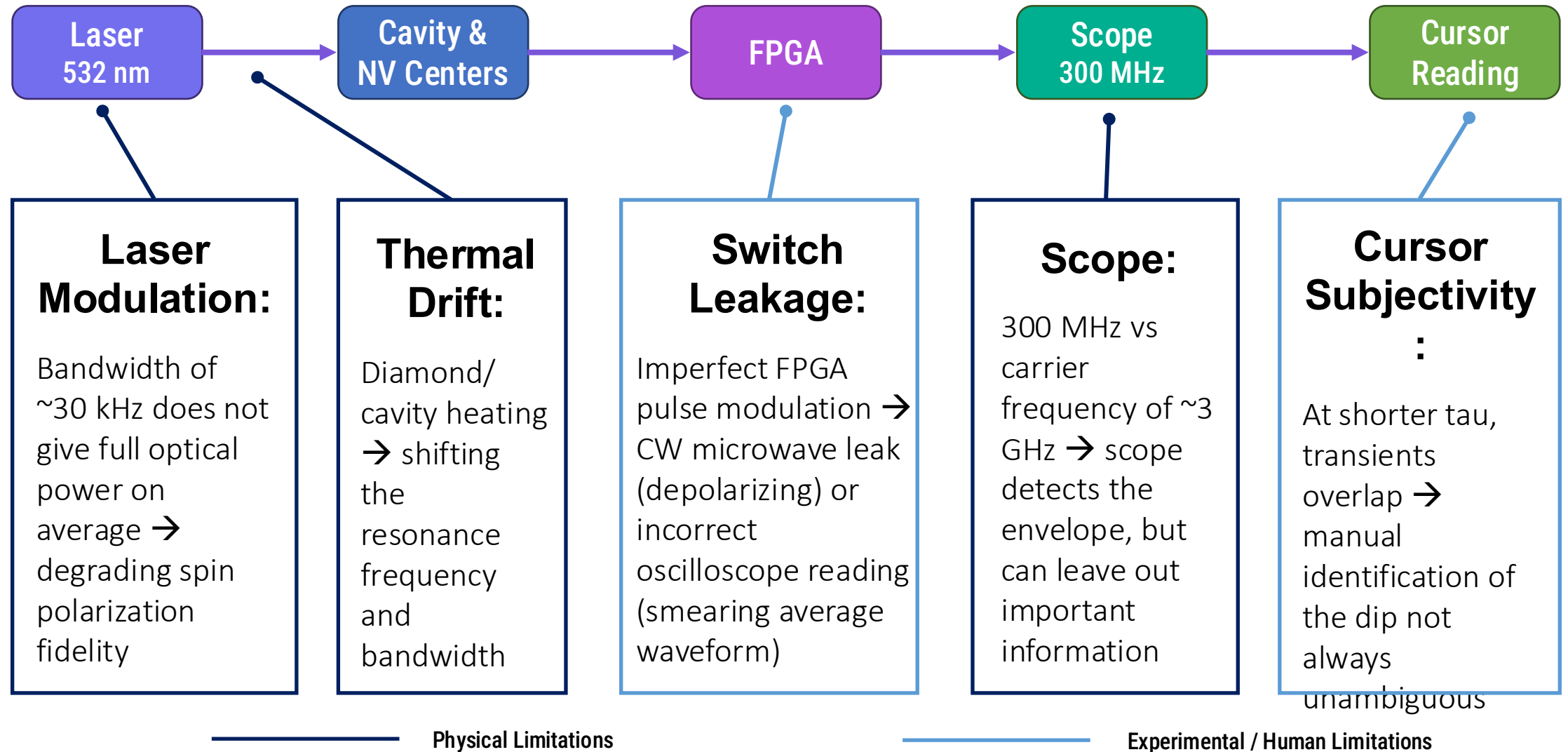


No clean Rabi oscillations resolvable in the tau sweep

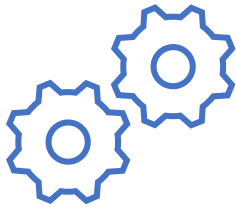


Curve fitting to $S(\tau)$ did not yet yield physically reliable Ω_R , T_R

Many stages in the measurement chain can introduce a distinct source of distortion or uncertainty

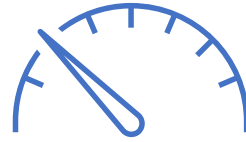


Resolving clean Rabi oscillations requires more averaging, higher drive power, and smarter FPGA control



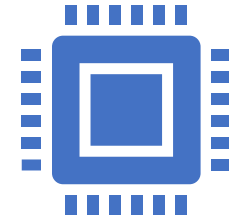
Hardware

- Optimize loop coupler position
- Replace scope with a fast digitizer
- Add an envelope detector
- Increase MW drive power
- Increase diamond NV density
- Add microwave power amplifier



Measurement

- Increase averaging beyond $N=512$
- Lock-in detection on the laser modulation
- 2D sweep with the VNA
- Replace manual cursor reading with automated waveform capture



FPGA

- Add a programmable tau register
- Add software-controlled trigger
- Implement on-FPGA averaging accumulator to compute trace averages without scope limits
- Readout gating with precise delay



We take a step toward room-temperature quantum sensing architectures



What we built

An FPGA-based pulsed control system for cavity-coupled NV magnetometry, with full integration of laser gating, microwave gating, and cavity readout



What we demonstrated

End-to-end pulsed operation of the cavity-coupled NV chain, yielding 13% spin-dependent contrast — confirming the signal originates from spin-cavity coupling rather than laser artifacts.



What we did not resolve

Clean Rabi oscillations remain buried at the noise floor of our single-shot scope readout, with coupling in the intermediate rather than strong regime.

- **Cavity-enhanced NV platforms** are promising for sensing-and-control architectures, with potential to enable compact, room-temperature quantum sensors for **brain imaging, materials characterization, and navigation**.
- Reaching that sensitivity requires **closing the measurement chain gaps** identified in our experiments.

Thank You!

We now welcome any questions about our project